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Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MECHATRONIC SYSTEMS
NQF LEVEL 2

27 February 2026

This marking guideline consists of 6 pages.

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QUESTION 1

- 1.1 1.1.1 Colour as an input is used to indicate to the operator or user what the function of the specific switching device operation is.✓✓(2 × 1) (2)
- 1.1.2 Colour as an output is used to enable operators or users of technical systems to identify the operating status of a machine, especially dangerous or malfunction situations.✓✓ (2 × 1) (2)
- 1.2 1.2.1 A bimetal is constructed by bonding two different metals✓ with distinct coefficients of thermal expansion,✓ meaning they expand at different rates when heated.✓ These metals are usually laminated or riveted together in strips.✓ (4 × 1) (4)
- 1.2.2
- Thermostats
 - Temperature sensors
 - Thermal Switches
 - Circuit breakers
 - Analogue thermometers
 - Electric irons or toasters
- (Any 4 × 1) (4)
- 1.3 1.3.1 When the bimetal is heated, the metal with the higher coefficient of thermal expansion (usually brass) expands more than the metal with the lower coefficient (usually steel).✓✓ (2 × 1) (2)
- 1.3.2 As the brass expands more than the steel, the bimetal bends or curves, causing the free end to move.✓✓ (2 × 1) (2)
- 1.3.3 When the bimetal is cooled, the process is reversed, and the bimetal returns to its original position.✓✓ (2 × 1) (2)
- 1.4
- Pulley system
 - Chain and sprocket system
 - Gear systems
- Any TWO✓✓ (2 × 1) (2)

[20]

QUESTION 2

- 2.1
- Capacitive proximity switch✓
 - Inductive proximity switch✓
 - Optical proximity switch✓
 - Ultrasonic proximity switch✓
 - Magnetic reed switch✓
- (5 × 1) (5)
- 2.2
- 2.2.1
- Optical proximity switches can detect objects with high accuracy.
 - Can detect objects at a distance of up to several meters.
 - Optical signals are not affected by electromagnetic interference.
 - Can detect transparent objects, such as glass or plastic.
- (4 × 1) (4)
- 2.2.2
- Dirt and dust can block the optical signal.
 - Ambient light can interfere with the optical signal.
 - The optical switch requires alignment with the object being detected.
- (3 × 1) (3)
- 2.3
- In mechatronic systems, computers are used in industry primarily because they enable precision,✓ automation,✓ and efficiency.✓
- (3 × 1) (3)
- 2.4
- High initial cost
 - Training Requirements
 - Dependence on technology
 - Limited creativity
 - Data management issues
 - Security concerns
 - Limited standardisation
 - Over-reliance on automation
 - Environmental impact
 - Potential for errors
- (Any 5 × 1) (5)

[20]

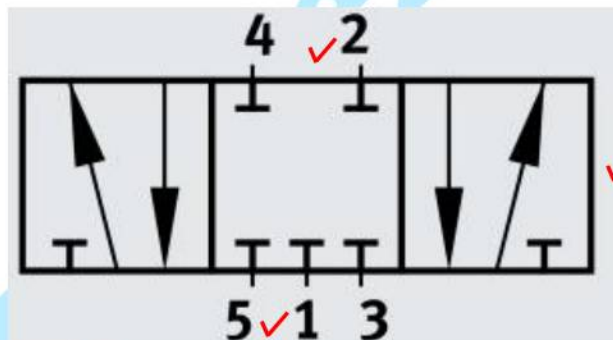
QUESTION 3

- 3.1.1 Ensure a reliable air supply with adequate pressure and flow rate.✓✓ (2 × 1) (2)
- 3.1.2 Install air filters to remove contaminants and moisture from the compressed air.✓✓ (2 × 1) (2)
- 3.1.3 Regulate air pressure to the recommended level for the pneumatic system.✓✓ (2 × 1) (2)
- 3.1.4 Lubricate pneumatic components, such as cylinders and valves, to reduce friction and wear.✓✓ (2 × 1) (2)

- 3.2 In FIGURE 1, port 1 is "blocked off" while ports 2 and 3 are connected,✓ when the spool is in the normal position, this position will allow air to exhaust via ports 2 and 3.✓✓
In FIGURE 2, when the 3/2-way valve is activated, the exhaust port 3 is "blocked off" by the spool, thus allowing air to flow via the valve from port 2 to 1.✓✓

(3 × 2) (6)

3.3



(3 × 1) (3)

- 3.4
- Warning lights.
 - Two-handed safety control
 - Pressure Relief Valves
 - Emergency Stop (E-Stop) Buttons
 - Pressure Sensors and Alarms

(Any 3 × 1) (3)

[20]

QUESTION 4

- 4.1 4.1.1
- Reliability
 - Easy signal processing
 - Low power consumption
 - Fast signal transmission
 - Flexibility
 - Long-Distance Control
 - Easy Integration with Electronic Devices
 - Precision and Accuracy

(Any relevant 5 × 1) (5)

- 4.1.2 Protection by earthing or grounding is a safety measure that protects people and equipment✓ from electrical shock and damage.✓ Earthing involves connecting electrical equipment, wiring, and systems to the earth, typically a grounded conductor.✓

(3 × 1) (3)

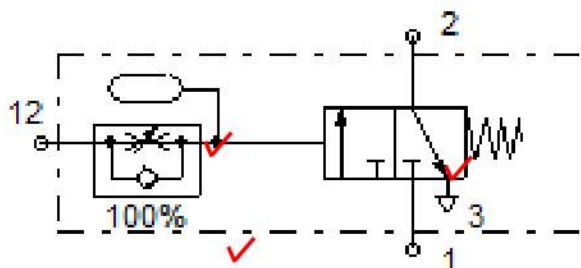
- 4.2
- Sequence control
 - Cycle counting
 - Production counting
 - Quality control
 - Time-based control
 - Packaging Machines
 - Automatic Drilling Machines
 - Part Sorting Systems
 - Press Machines (Punching or Stamping)

(Any relevant 4 × 1) (4)

- 4.3 Once the timer times out,✓ it will cause the relay K1 to de-energise,✓ thus removing the supply from the coil of 1M1.✓ The spring will then return the valve to its normal position,✓ thus causing the cylinder to retract.✓

(5 × 1) (5)

4.4



(3 × 1) (3)

[20]

QUESTION 5

5.1

Fault Finding	Fault Diagnosis
Focuses on identifying and isolating the fault.	Fault diagnosis focuses on analysing and understanding the root cause.
Involves a narrower scope, focusing on a specific symptom or component.	Involves a broader scope, considering the entire system and its interactions.
Often relies on observational and deductive techniques.	Involves more analytical and scientific methods, such as data analysis and experimentation.

(Any relevant 3 × 2)

(6)

5.2

- Check the seals of the connectors for leaks
- Replace lines connected to moving parts
- Check the rod bearings in the cylinders for wear and replace if necessary
- Check the function of safety valves
- Check mountings
- Inspect and Replace Air Filters
- Check and Clean Air Dryers
- Test System Pressure Settings
- Lubrication of Moving Parts

(Any relevant 6 × 1)

(6)

5.3

- Wear PPE, such as safety glasses and gloves, to prevent injury.✓
- Ensure the circuit is de-energised before performing continuity testing.✓
- Use insulated tools to prevent electrical shock.✓

(Any relevant 3 × 1)

(3)

5.4

- Before removing tubing, ensure the pneumatic circuit is depressurised to prevent injury from escaping air.
- Disconnect power to the pneumatic system to prevent accidental start-up.
- Ensure safe disposal of the removed tubing and any associated components.

(Any relevant 2 × 1)

(2)

5.5

To protect a circuit from overload current, you would use protective devices✓ that automatically interrupt the power supply ✓when the current exceeds safe levels.✓

(3 × 1)

(3)

[20]

TOTAL:

100